

Qualification Pack



Ceramics Preparation Artisan

Electives: Casting Operator- Ceramics/ Colour Mixing/ Engobing Operator (Ceramics)/ Furnace Operator/ Glazing Operator/ Jigger Operator/ Moulder/ Mixing Operator/ Perform Slip Trailing

QP Code: HCS/Q0801

Version: 3.0

NSQF Level: 2.5

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HCS/Q0801: Ceramics Preparation Artisan

Brief Job Description

Ceramics Preparation Artisan prepares ceramics in various steps and various methods to make final prepared ceramic for further processes.

Personal Attributes

A good Ceramics Preparation Artisan is detail-oriented and has a strong sense of creativity, along with good hands-on skills for working with clay. They are patient and organised, ensuring that materials are prepared correctly for the best results.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [HCS/N0110: Packaging and Presentation of Ceramic Products](#)
2. [HCS/N8509: Entrepreneurship Skill for Ceramics Preparation Artisan](#)
3. [HCS/N9952: Working together as a team](#)
4. [HCS/N9946: Managing workplace safety](#)
5. [HCS/N9937: Maintain Cleanliness](#)
6. [HCS/N9941: Understanding Business Operations](#)
7. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

Electives(*mandatory to select at least one*):

Elective 1: Casting Operator- Ceramics

The Casting Operator in ceramics is responsible for pouring liquid clay into moulds to create various ceramic shapes and ensuring that the casting process runs smoothly. This role requires attention to detail and precision to achieve high-quality finished products.

1. [HCS/N0602: Carry out the processes required to prepare the cast](#)

Elective 2: Colour Mixing

Colour mixing in ceramics is about blending different colours and glazes to make new shades. It requires knowing how colours work together to get the right look, so the finished pieces are bright and even after they are fired in the kiln.

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1. [HCS/N0816: Mix the ingredients to obtain the desired colours](#)

Elective 3: Engobing Operator (Ceramics)

An Engobing Operator in ceramics applies a layer of engobe, which is a liquid clay mixture, to the surface of ceramic pieces. This process helps to add colour, texture, and a smooth finish to the product before it is fired.

1. [HCS/N0703: Engobing of ceramics products](#)

Elective 4: Furnace Operator

A Furnace Operator in ceramics is responsible for managing and operating kilns used to fire ceramic pieces.

1. [HCS/N2104: Carry out furnace operation](#)

Elective 5: Glazing Operator

A Glazing Operator in ceramics applies glaze to finished pieces to enhance their appearance and durability. This role involves preparing the glaze, ensuring even application, and sometimes adjusting the glaze mixture for desired effects.

1. [HCS/N0817: Perform Glazing of Ceramics Products](#)

Elective 6: Jigger Operator

A Jigger Operator in ceramics is responsible for using a jigger machine to shape and form clay into specific designs.

1. [HCS/N0502: Perform jiggering of Dough](#)

Elective 7: Moulder

A Moulder in ceramics is responsible for shaping and forming clay into specific designs, often using hand techniques or molds.

1. [HCS/N0408: Prepare the Mould for production](#)

Elective 8: Mixing Operator

A Mixing Operator in ceramics is responsible for preparing and mixing raw materials, such as clay, glazes, and other additives, to create the desired consistency and properties for ceramic production.

1. [HCS/N0303: Perform mixing of raw materials using equipment](#)

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2. [HCS/N0304: Perform Manual mixing](#)

Elective 9: Perform Slip Trailing

Slip trailing is a decorative technique used in ceramics where a liquid clay mixture, known as slip, is applied to the surface of a ceramic piece to create intricate designs and patterns.

1. [HCS/N0109: Perform Slip Trailing](#)

Qualification Pack (QP) Parameters

Sector	Handicrafts and Carpet
Sub-Sector	Ceramics
Occupation	Pre-Crafting
Country	India
NSQF Level	2.5
Credits	10
Aligned to NCO/ISCO/ISIC Code	NCO-2015/7314.0400
Minimum Educational Qualification & Experience	9th grade pass with NA of experience OR 5th grade pass with 2 Years of experience Relevant experience OR Ability to read and write with 4 Years of experience Relevant experience OR Previous relevant Qualification of NSQF Level 2 with 6 Months of experience Relevant experience
Minimum Level of Education for Training in School	Ability to read and write
Pre-Requisite License or Training	NA
Minimum Job Entry Age	14 Years
Last Reviewed On	NA
Next Review Date	30/04/2028
NSQC Approval Date	30/04/2025

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Version	3.0
Reference code on NQR	QG-2.5-HC-04004-2025-V2-HCSSC
NQR Version	3

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HCS/N0110: Packaging and Presentation of Ceramic Products

Description

This unit focuses on the packaging and presentation of ceramic products, emphasizing the importance of protective measures, branding, and aesthetic appeal. It covers the selection of materials, packing techniques, and the use of eco-friendly options.

Scope

The scope covers the following :

- packaging for protection
- presentation for aesthetic appeal
- storytelling and experience
- sustainability considerations
- point of sale presentation

Elements and Performance Criteria

Packaging for Protection

To be competent, the user/individual on the job must be able to:

- PC1.** Use the bubble wrap and foam materials to cushion the ceramic items and protect them from shocks and impacts during transportation
- PC2.** use corrugated cardboard boxes provide structural integrity
- PC3.** fill any empty spaces within the box to prevent movement during transit
- PC4.** use cardboard dividers for fragile items (like plates or bowls) to prevent them from touching each other
- PC5.** tape the boxes and consider using fragile stickers or labels to indicate care needed when handling

Presentation for Aesthetic Appeal

To be competent, the user/individual on the job must be able to:

- PC6.** invest in branded boxes or wrapping paper featuring the logo, which can enhance brand recognition and create a professional image
- PC7.** include labels with product information, care instructions, and brand storytelling
- PC8.** use colors that complement the ceramics, enhancing the overall aesthetic
- PC9.** a minimalist white box might highlight a colorful ceramic piece
- PC10.** use boxes with transparent windows to showcase the product without opening the packaging

Storytelling and Experience

To be competent, the user/individual on the job must be able to:

- PC11.** share the story behind the ceramics products
- PC12.** add a small surprise element, like a thank-you note or a sample of another product

Sustainability Considerations

To be competent, the user/individual on the job must be able to:

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PC13. opt for biodegradable or recyclable packaging materials to appeal to environmentally-conscious consumers

PC14. reduce excess packaging can not only lower costs but also align with sustainable practices

Point of Sale Presentation

To be competent, the user/individual on the job must be able to:

PC15. create displays that tell a story or showcase a collection (e.g., seasonal themes, colors, or uses) to attract customer interest

PC16. give information about the artisans, the ceramics-making process and the unique qualities of the products

PC17. offer ideas on how to display or use the ceramic products, such as pairing them with certain decor styles or recommending complementary products

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. knowledge of different packaging materials including cardboard boxes, foam inserts, bubble wrap, or biodegradable materials

KU2. understand the types of display stands, fabric wraps, or boxes used for presentation can enhance the visual appeal of ceramic products

KU3. ceramics are often fragile, so the packaging must effectively protect against impact, vibration, and temperature changes

KU4. knowing how ceramics can be safely stacked and transported is essential to prevent breakage during shipping and handling

KU5. knowledge of graphic design can help create visually appealing packaging that aligns with the overall brand image

KU6. understand the color theory and design principles can enhance the presentation

KU7. keeping up with market trends, such as sustainable packaging or minimalistic designs, can inform packaging choices

KU8. knowledge of safety standards for packaging, especially for food-safe ceramics, ensures that products meet health regulations

KU9. packaging can communicate the story of the product or the artist

KU10. knowledge of size and weight helps in choosing packaging that minimizes shipping costs while ensuring the product is well-protected

KU11. packaging should include clear information about the product, such as care instructions, materials, and dimensions

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. ensure all elements of the packaging and presentation are carefully considered and executed, from labeling to the arrangement of products

GS2. develop innovative packaging designs and presentation strategies that stand out in a competitive market and resonate with target customers

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- GS3.** effectively conveying information about the products and the brand story through packaging and marketing materials, both visually and textually
- GS4.** properties of different packaging materials includes knowing how well they protect ceramics and their environmental impact
- GS5.** clear labeling and messaging are essential for effective presentation. Good communication ensures that customers understand product details and care instructions

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Packaging for Protection</i>	9	20	-	-
PC1. Use the bubble wrap and foam materials to cushion the ceramic items and protect them from shocks and impacts during transportation	2	4	-	-
PC2. use corrugated cardboard boxes provide structural integrity	2	4	-	-
PC3. fill any empty spaces within the box to prevent movement during transit	2	4	-	-
PC4. use cardboard dividers for fragile items (like plates or bowls) to prevent them from touching each other	2	4	-	-
PC5. tape the boxes and consider using fragile stickers or labels to indicate care needed when handling	1	4	-	-
<i>Presentation for Aesthetic Appeal</i>	10	20	-	-
PC6. invest in branded boxes or wrapping paper featuring the logo, which can enhance brand recognition and create a professional image	2	4	-	-
PC7. include labels with product information, care instructions, and brand storytelling	2	4	-	-
PC8. use colors that complement the ceramics, enhancing the overall aesthetic	2	4	-	-
PC9. a minimalist white box might highlight a colorful ceramic piece	2	4	-	-
PC10. use boxes with transparent windows to showcase the product without opening the packaging	2	4	-	-
<i>Storytelling and Experience</i>	3	10	-	-
PC11. share the story behind the ceramics products	2	6	-	-
PC12. add a small surprise element, like a thank-you note or a sample of another product	1	4	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Sustainability Considerations</i>	4	8	-	-
PC13. opt for biodegradable or recyclable packaging materials to appeal to environmentally-conscious consumers	2	4	-	-
PC14. reduce excess packaging can not only lower costs but also align with sustainable practices	2	4	-	-
<i>Point of Sale Presentation</i>	4	12	-	-
PC15. create displays that tell a story or showcase a collection (e.g., seasonal themes, colors, or uses) to attract customer interest	2	4	-	-
PC16. give information about the artisans, the ceramics-making process and the unique qualities of the products	1	4	-	-
PC17. offer ideas on how to display or use the ceramic products, such as pairing them with certain decor styles or recommending complementary products	1	4	-	-
NOS Total	30	70	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	HCS/N0110
NOS Name	Packaging and Presentation of Ceramic Products
Sector	Handicrafts and Carpet
Sub-Sector	
Occupation	Production Management-Ceramics
NSQF Level	2.5
Credits	2
Version	1.0
Last Reviewed Date	30/04/2025
Next Review Date	30/04/2028
NSQC Clearance Date	30/04/2025

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HCS/N8509: Entrepreneurship Skill for Ceramics Preparation Artisan

Description

This unit is designed to equip ceramics preparation artisans with the essential entrepreneurship skills needed to build, manage, and grow a successful business in the ceramics industry. It focuses on providing the practical knowledge and abilities required to operate effectively as a creative entrepreneur, from product development to financial management and marketing.

Scope

The scope covers the following :

- Introduction to Entrepreneurship
- Entrepreneurial mindset
- Business planning for ceramics artisans
- Marketing and branding
- Financial management
- Sales and customer relations
- Networking and collaboration
- Problem-solving and adaptability

Elements and Performance Criteria

Introduction to Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC1.** Overview of entrepreneurship and its relevance to ceramics artisans.
- PC2.** Characteristics of successful entrepreneur artisans.

Entrepreneurial Mindset

To be competent, the user/individual on the job must be able to:

- PC3.** Learn creativity and innovation in the context of ceramics.
- PC4.** Embracing new ideas and being willing to experiment with different approaches to problem-solving.

Business Planning for Ceramics Artisans

To be competent, the user/individual on the job must be able to:

- PC5.** Craft a comprehensive business plan.
- PC6.** Use market research method and identifying the niche (luxury ceramics, functional pottery, decorative pieces).
- PC7.** Define a clear business goals and timelines for growth.

Marketing and Branding

To be competent, the user/individual on the job must be able to:

- PC8.** Develop a unique brand identity for ceramics products.
- PC9.** Use storytelling and behind-the-scenes content to engage customers.
- PC10.** Identify the social media and online platforms to showcase ceramics work.

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Financial Management

To be competent, the user/individual on the job must be able to:

- PC11.** Manage production costs (materials, tools, kiln usage).
- PC12.** Use simple financial management software or even spreadsheets to track cash flow, profits, and losses.

Sales and Customer Relations

To be competent, the user/individual on the job must be able to:

- PC13.** Use effective sales techniques for artisans.
- PC14.** Build a lasting customer relationships through custom orders and feedback.

Networking and Collaboration

To be competent, the user/individual on the job must be able to:

- PC15.** Importance of building connections with other artisans, suppliers, and retailers.
- PC16.** Opportunities for collaborations with other artists or designers.
- PC17.** Join artisan associations, participate in workshops, or visit pottery studios to meet like-minded individuals and potential business partners.

Problem-Solving and Adaptability

To be competent, the user/individual on the job must be able to:

- PC18.** Handling production challenges like kiln issues or material shortages.
- PC19.** Adapting designs and production methods based on market trends or customer preferences.
- PC20.** Manage time effectively to balance production, marketing, and business management.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** know the craft deeply, including material properties (clay types, glaze chemistry, firing techniques), and the needs of the target customers
- KU2.** different types of ceramics (e.g., earthenware, stoneware, porcelain) and their uses
- KU3.** preferences and trends in the ceramics market, such as minimalistic styles, functional home decor, or artistic sculptures
- KU4.** pricing strategies based on product quality, uniqueness, and customer expectations
- KU5.** understand how to create unique selling propositions (USP) by focusing on design innovations, sustainable practices, or craftsmanship techniques
- KU6.** study successful artisans and identify gaps or opportunities in the market to position the brand uniquely
- KU7.** understand batch production versus custom-made products and when to apply each based on customer demand and pricing
- KU8.** understand the cost structure, including fixed costs (e.g., equipment, rent) and variable costs (e.g., materials, labor)
- KU9.** know how to interact with customers professionally, both online and offline, and respond to feedback or complaints constructively
- KU10.** develop loyalty programs or after-sales services to encourage repeat purchases

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- KU11.** understand local business registration requirements, health and safety regulations for the workshops
- KU12.** identify potential risks such as supply chain issues, kiln failures, or market downturns and have contingency plans in place
- KU13.** learn sustainable practices in ceramics, such as minimizing waste, using non-toxic glazes, recycling water in the workshop, or firing in energy-efficient kilns

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** write down the concept of entrepreneurship and their characteristics
- GS2.** interact with employees to work efficiently
- GS3.** identify trends and customer preferences
- GS4.** do surveys or analyze social media trends to find out what styles of ceramics are currently popular
- GS5.** identify the fundamentals of running a business, including finance, marketing, and sales
- GS6.** detect problems in day-to-day tasks
- GS7.** ability to identify and address challenges that arise in the creative process or business operations
- GS8.** keep track of material costs and labor hours to ensure profitability while identifying potential areas for cost reduction without compromising quality
- GS9.** able to tell the story behind a piece, its inspiration, and the craftsmanship involved can make it more appealing to potential buyers
- GS10.** finding ways to reduce the occurrence of cracks in pottery during the drying or firing process

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Entrepreneurship</i>	4	6	-	-
PC1. Overview of entrepreneurship and its relevance to ceramics artisans.	2	3	-	-
PC2. Characteristics of successful entrepreneur artisans.	2	3	-	-
<i>Entrepreneurial Mindset</i>	4	6	-	-
PC3. Learn creativity and innovation in the context of ceramics.	2	3	-	-
PC4. Embracing new ideas and being willing to experiment with different approaches to problem-solving.	2	3	-	-
<i>Business Planning for Ceramics Artisans</i>	6	9	-	-
PC5. Craft a comprehensive business plan.	2	3	-	-
PC6. Use market research method and identifying the niche (luxury ceramics, functional pottery, decorative pieces).	2	3	-	-
PC7. Define a clear business goals and timelines for growth.	2	3	-	-
<i>Marketing and Branding</i>	6	9	-	-
PC8. Develop a unique brand identity for ceramics products.	2	3	-	-
PC9. Use storytelling and behind-the-scenes content to engage customers.	2	3	-	-
PC10. Identify the social media and online platforms to showcase ceramics work.	2	3	-	-
<i>Financial Management</i>	4	6	-	-
PC11. Manage production costs (materials, tools, kiln usage).	2	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. Use simple financial management software or even spreadsheets to track cash flow, profits, and losses.	2	3	-	-
<i>Sales and Customer Relations</i>	4	6	-	-
PC13. Use effective sales techniques for artisans.	2	3	-	-
PC14. Build a lasting customer relationships through custom orders and feedback.	2	3	-	-
<i>Networking and Collaboration</i>	6	9	-	-
PC15. Importance of building connections with other artisans, suppliers, and retailers.	2	3	-	-
PC16. Opportunities for collaborations with other artists or designers.	2	3	-	-
PC17. Join artisan associations, participate in workshops, or visit pottery studios to meet like-minded individuals and potential business partners.	2	3	-	-
<i>Problem-Solving and Adaptability</i>	6	9	-	-
PC18. Handling production challenges like kiln issues or material shortages.	2	3	-	-
PC19. Adapting designs and production methods based on market trends or customer preferences.	2	3	-	-
PC20. Manage time effectively to balance production, marketing, and business management.	2	3	-	-
NOS Total	40	60	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	HCS/N8509
NOS Name	Entrepreneurship Skill for Ceramics Preparation Artisan
Sector	Handicrafts and Carpet
Sub-Sector	
Occupation	Entrepreneur, Product R&D - Ceramics
NSQF Level	2.5
Credits	1
Version	1.0
Last Reviewed Date	30/04/2025
Next Review Date	30/04/2028
NSQC Clearance Date	30/04/2025

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HCS/N9952: Working together as a team

Description

This unit is about working as part of a team in the process.

Scope

The scope covers the following :

- commitment and trust
- communication
- adaptability
- creative freedom

Elements and Performance Criteria

Commitment and trust

To be competent, the user/individual on the job must be able to:

- PC1.** be accountable to one's role in the whole process
- PC2.** perform all roles with full

Communication

To be competent, the user/individual on the job must be able to:

- PC3.** report problems faced during the process
- PC4.** talk politely with other team members and colleagues
- PC5.** submit a daily report of own performance

Adaptability

To be competent, the user/individual on the job must be able to:

- PC6.** adjust in different work situations
- PC7.** give due importance to other's point of view
- PC8.** avoid conflicting situations

Creative freedom

To be competent, the user/individual on the job must be able to:

- PC9.** develop new ideas for work procedures
- PC10.** improve upon the existing techniques to increase process efficiency

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** general rules and regulations in the sector
- KU2.** procedure followed to get the final output
- KU3.** safe working practices to be adopted
- KU4.** to report the supervisor or higher authority about any grievances faced

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- KU5.** understanding the importance of the previous and next step of the process
- KU6.** process flow in the handicraft section
- KU7.** material sequence of flow
- KU8.** functions of different parts of product development
- KU9.** tools and equipment used
- KU10.** guidelines for operating the equipment
- KU11.** safety procedures to be followed as applicable

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** prepare status and progress reports
- GS2.** write memos and e-mail to co-workers, and vendors to provide them with work updates and to request appropriate information without English language errors regarding grammar or sentence construct
- GS3.** keep abreast with the latest knowledge by reading brochures, pamphlets, and product information sheets
- GS4.** discuss task lists, schedules, and work-loads with co-workers
- GS5.** keep co-workers and supervisors informed about progress
- GS6.** make decisions pertaining to the concerned area of work
- GS7.** plan and organize service feedback files/documents
- GS8.** manage relationships with customers
- GS9.** build customer relationships and use customer centric approach
- GS10.**
 - think through the problem, evaluate the possible solution(s) and suggest an optimum /best
 - possible solution(s)
- GS11.** identify immediate or temporary solutions to resolve delays
- GS12.** use the existing data to arrive at specific data points
- GS13.**
 - apply, analyze, and evaluate the information gathered from observation, experience,
 - reasoning, or communication, as a guide to thought and action

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Commitment and trust</i>	6	14	-	-
PC1. be accountable to one's role in the whole process	3	7	-	-
PC2. perform all roles with full	3	7	-	-
<i>Communication</i>	9	21	-	-
PC3. report problems faced during the process	3	7	-	-
PC4. talk politely with other team members and colleagues	3	7	-	-
PC5. submit a daily report of own performance	3	7	-	-
<i>Adaptability</i>	9	21	-	-
PC6. adjust in different work situations	3	7	-	-
PC7. give due importance to other's point of view	3	7	-	-
PC8. avoid conflicting situations	3	7	-	-
<i>Creative freedom</i>	6	14	-	-
PC9. develop new ideas for work procedures	3	7	-	-
PC10. improve upon the existing techniques to increase process efficiency	3	7	-	-
NOS Total	30	70	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	HCS/N9952
NOS Name	Working together as a team
Sector	Handicrafts and Carpet
Sub-Sector	
Occupation	Generic Handicrafts and Carpet
NSQF Level	2.5
Credits	1
Version	1.0
Last Reviewed Date	30/04/2025
Next Review Date	30/04/2028
NSQC Clearance Date	30/04/2025

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HCS/N9946: Managing workplace safety

Description

This unit provides Performance Criteria, Knowledge & Understanding, and Skills & Abilities required to comply with health, safety, and security requirements at the workplace and covers procedures to prevent, control, and minimize risk to self and others.

Scope

The scope covers the following :

- Comply with health, safety, and security requirements at work

Elements and Performance Criteria

Comply with health, safety, and security requirements at work

To be competent, the user/individual on the job must be able to:

- PC1.** comply with health and safety related instructions applicable to the workplace.
- PC2.** use and maintain personal protective equipment as per protocol.
- PC3.** carry out own activities in line with approved guidelines and procedures
- PC4.** maintain a healthy lifestyle and guard against dependency on intoxicants.
- PC5.** follow environment management system related procedures.
- PC6.** store materials and tools in line with manufacturers and organisational requirements
- PC7.** safely handle and move waste and debris.
- PC8.** minimize health and safety risks to self and others due to own actions
- PC9.** seek clarifications, from supervisors or other authorized personnel in case of perceived risks
- PC10.** monitor the workplace and work processes for potential risks and threats.
- PC11.** carry out periodic walk-through to keep work area free from hazards and obstructions, if assigned
- PC12.** report hazards and potential risks/ threats to supervisors or other authorized personnel
- PC13.** participate in mock drills/ evacuation procedures organized at the workplace
- PC14.** undertake first aid, fire-fighting and emergency response training, if asked to do so
- PC15.** take action based on instructions in the event of fire, emergencies or accidents
- PC16.** follow organisation procedures for evacuation when required

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** health and safety related practices applicable at the workplace
- KU2.** potential hazards, risks and threats based on nature of operations
- KU3.** organizational procedures for safe handling of tools

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- KU4.** potential risks due to own actions and methods to minimize these
- KU5.** environmental management system related procedures at the workplace
- KU6.** layout of the plant and details of emergency exits, escape routes, emergency equipment and assembly points
- KU7.** potential accidents and emergencies and response to these scenarios
- KU8.** reporting protocol and documentation required
- KU9.** details of personnel trained in first aid, fire-fighting and emergency response
- KU10.** actions to take in the event of a mock drills/ evacuation procedures or actual accident, emergency or fire.
- KU11.** occupational health and safety risks and methods
- KU12.** personal protective equipment and method of use
- KU13.** identification, handling and storage of hazardous substances
- KU14.** proper disposal system for waste and by-products
- KU15.** signage related to health and safety and their meaning
- KU16.** importance of sound health, hygiene and good habits
- KU17.** ill-effects of alcohol, tobacco and drugs

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** write in local language.
- GS2.** read measurement instructions
- GS3.** communicate orally with colleagues
- GS4.** follow organization rule-based decision making process
- GS5.** take decision with systematic course of actions and/or response
- GS6.** plan and organize your work to achieve targets and deadlines
- GS7.** manage relationships with customers
- GS8.** build customer relationships and use customer centric approach.
- GS9.** think through the problem, evaluate the possible solution(s) and suggest an optimum /best possible solution(s)
- GS10.** identify immediate or temporary solutions to resolve delays
- GS11.** analyze data and activities.
- GS12.** pass on relevant information to others
- GS13.** apply, analyze, and evaluate the information gathered from observation, experience, reasoning, or communication, as a guide to thought and action

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Comply with health, safety, and security requirements at work</i>	30	70	-	-
PC1. comply with health and safety related instructions applicable to the workplace.	2	6	-	-
PC2. use and maintain personal protective equipment as per protocol.	2	6	-	-
PC3. carry out own activities in line with approved guidelines and procedures	2	6	-	-
PC4. maintain a healthy lifestyle and guard against dependency on intoxicants.	2	6	-	-
PC5. follow environment management system related procedures.	2	4	-	-
PC6. store materials and tools in line with manufacturers and organisational requirements	2	3	-	-
PC7. safely handle and move waste and debris.	1	3	-	-
PC8. minimize health and safety risks to self and others due to own actions	2	4	-	-
PC9. seek clarifications, from supervisors or other authorized personnel in case of perceived risks	1	3	-	-
PC10. monitor the workplace and work processes for potential risks and threats.	1	3	-	-
PC11. carry out periodic walk-through to keep work area free from hazards and obstructions, if assigned	2	3	-	-
PC12. report hazards and potential risks/ threats to supervisors or other authorized personnel	3	4	-	-
PC13. participate in mock drills/ evacuation procedures organized at the workplace	2	3	-	-
PC14. undertake first aid, fire-fighting and emergency response training, if asked to do so	2	4	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC15. take action based on instructions in the event of fire, emergencies or accidents	2	6	-	-
PC16. follow organisation procedures for evacuation when required	2	6	-	-
NOS Total	30	70	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	HCS/N9946
NOS Name	Managing workplace safety
Sector	Handicrafts and Carpet
Sub-Sector	
Occupation	Generic Handicrafts and Carpet
NSQF Level	2.5
Credits	1
Version	1.0
Last Reviewed Date	30/04/2025
Next Review Date	30/04/2028
NSQC Clearance Date	30/04/2025

Qualification Pack

HCS/N9937: Maintain Cleanliness

Description

This NOS provides the abilities required for taking responsibility for their own health at the workplace and is about using the correct procedures to prevent, control and minimize risk to them and others at the workplace.

Scope

The scope covers the following :

- adopt healthy work practices
- achieve work productivity while maintaining health

Elements and Performance Criteria

Adopt healthy work practices

To be competent, the user/individual on the job must be able to:

- PC1.** always cover the mouth and nose with a dust mask while working and keep on changing when it gets blocked with dust
- PC2.** follow work instructions strictly to reduce the amount of pollution at the work place e.g. wet the rock / craft material before working on it
- PC3.** wear protective goggles over eyes and replace them when scratches on it obscure the vision
- PC4.** wear gloves as per the materials used for making handicraft to avoid blisters; scratches and cuts
- PC5.** undergo preventive health checkups at regular intervals
- PC6.** take prompt treatment from the doctor in case of illness

Achieve work productivity while maintaining health

To be competent, the user/individual on the job must be able to:

- PC7.** follow SOPs for dealing with blisters; scratches; accidental fires or any other type of emergencies at work
- PC8.** ensure the absence of no productivity loss or absenteeism from work due to illness
- PC9.** ensure no long term ill effect on the personal health

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** companys policies on personal health and occupational hazard management
- KU2.** companys hr policies
- KU3.** companys reporting structure
- KU4.** companys emergency evacuation procedure
- KU5.** health risks to the worker at the work place

Qualification Pack

- KU6.** healthy work practices
- KU7.** how to perform the duties in a way to minimize pollution at the work place
- KU8.** what personal protective equipments should be worn and how it is cared for
- KU9.** safe disposal methods for waste
- KU10.** how to provide the first aid treatment at workplace
- KU11.** emergency procedures to be followed in case of an mishap such as fire accidents etc.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read personal health instructions and manual
- GS2.** read the usage of various hand tools and personal protection equipments
- GS3.** take notes on descriptions and details of various personal health maintenance procedures
- GS4.** communicate with supervisor about the physical symptoms
- GS5.** receive instructions from doctor and supervisor on medical care
- GS6.** how to select appropriate hand tools and personal protection equipment
- GS7.** when to change personal protection equipment during work
- GS8.** how to identify first aid needs in case and of an injury
- GS9.** how to select appropriate hand tools and personal protection equipment
- GS10.** when to change personal protection equipment during work
- GS11.** how to use materials that does not affect customer health / make injury
- GS12.** improve work processes by adopting best safety practices
- GS13.** analyze the usage of appropriate tools and consumables
- GS14.** spot errors and any other disruptions and communicate with solutions

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Adopt healthy work practices</i>	18	48	-	-
PC1. always cover the mouth and nose with a dust mask while working and keep on changing when it gets blocked with dust	3	8	-	-
PC2. follow work instructions strictly to reduce the amount of pollution at the work place e.g. wet the rock / craft material before working on it	3	8	-	-
PC3. wear protective goggles over eyes and replace them when scratches on it obscure the vision	3	8	-	-
PC4. wear gloves as per the materials used for making handicraft to avoid blisters; scratches and cuts	3	8	-	-
PC5. undergo preventive health checkups at regular intervals	3	8	-	-
PC6. take prompt treatment from the doctor in case of illness	3	8	-	-
<i>Achieve work productivity while maintaining health</i>	12	22	-	-
PC7. follow SOPs for dealing with blisters; scratches; accidental fires or any other type of emergencies at work	4	7	-	-
PC8. ensure the absence of no productivity loss or absenteeism from work due to illness	4	7	-	-
PC9. ensure no long term ill effect on the personal health	4	8	-	-
NOS Total	30	70	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	HCS/N9937
NOS Name	Maintain Cleanliness
Sector	Handicrafts and Carpet
Sub-Sector	Handicrafts (Ceramics), Handicrafts (Fashion Jewellery), Handicrafts (Stonecraft), Glassware, Metalware
Occupation	Generic Handicrafts and Carpet
NSQF Level	2.5
Credits	1
Version	1.0
Last Reviewed Date	30/04/2025
Next Review Date	30/04/2028
NSQC Clearance Date	30/04/2025

Qualification Pack

HCS/N9941: Understanding Business Operations

Description

This OS unit focuses on business planning, internal strategy, timelines.

Scope

The scope covers the following :

- people management
- product planning
- procurement of raw materials
- market interfacing
- financial management
- record keeping

Elements and Performance Criteria

People management

To be competent, the user/individual on the job must be able to:

- PC1.** arrange Interactive meetings of managers of sales and production teams and categorize the issues and feedbacks of both the teams
- PC2.** train the employees of his/her unit with the appropriate skills required to make market relevant and quality products
- PC3.** motivate the employees

Product planning

To be competent, the user/individual on the job must be able to:

- PC4.** compile a report based on old production reports
- PC5.** address the issues faced in previous productions and try to resolve them
- PC6.** gather and analyze the cues from the market
- PC7.** ascertain the customer preference
- PC8.** develop product range lines based on current market preference
- PC9.** develop product range lines that are unique and able to price high
- PC10.** price the products according to market trends
- PC11.** identify the competent marketing strategy for the product range

Procurement of raw materials

To be competent, the user/individual on the job must be able to:

- PC12.** list of the raw material s and prepare a B.O.M according to the product lines
- PC13.** ascertain the quantity and right price to procure the materials
- PC14.** identify the right locations/agents from where the raw materials can be procured
- PC15.** negotiate to get the best price
- PC16.** ensure quality materials are procured

Qualification Pack

PC17. ensure the procured materials are stored in appropriate conditions

PC18. compile a record of price quotations, POs, and bills of procurement for future reference

Market interfacing

To be competent, the user/individual on the job must be able to:

PC19. maintain a healthy and professional relationship with vendor

PC20. the competitive market falls in order with the company policies of best price, quality, and delivery parameters

PC21. analyze the prevalent price for product lines

PC22. decide on the most effective means to access the market

PC23. plan for cost-effective transportation to the market

PC24. position the product according to market requirements

PC25. identify and address the expectations of customer

Financial management

To be competent, the user/individual on the job must be able to:

PC26. analyze and ascertain the cost of production

PC27. maintain the book of accounts related to the business

PC28. maintain export documents like a letter of credit, custom clearance

PC29. identify cost-effective means of running the business

Record keeping

To be competent, the user/individual on the job must be able to:

PC30. identify various aspects of business that require recording

PC31. design formats for recording

PC32. compile various records of all aspects of the business

PC33. maintain these records with periodic updations

PC34. maintain necessary documents as per local government and regulatory requirements

PC35. reframe the procurement strategy according to local scenarios like weather conditions, transport strikes, affected prices, etc.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. interpersonal skills and communication with a cross-section of stakeholders

KU2. understanding of basics accounting so that he/she can monitor the transactions between company and vendor

KU3. knowledge of banking basics to keep the export process flawless

KU4. understanding of costing principles to calculate the cost of production with all the overheads and the actual price

KU5. product and craft knowledge including material and tools requirements

KU6. gathering market intelligence

KU7. various transportation means and implications on costing

KU8. various product lines that can be created depending on the sector of operation

Qualification Pack

- KU9.** basic record-keeping techniques
- KU10.** basic laws, rules, regulations, etc. regarding business
- KU11.** vendor management and development
- KU12.** pricing techniques
- KU13.** business profitability assessment

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** analyse product range and keep abreast of market trends
- GS2.** document various aspects of the business
- GS3.** compile descriptions and details about investment, expenditures, and sales
- GS4.** interact with teams to work efficiently
- GS5.** communicate and manage vendors for cost-effective outsourcing and procurements
- GS6.** interface with fellow entrepreneurs to exchange ideas on the business
- GS7.** communicate with the customers for their feedback about the product
- GS8.** comprehend information shared by various stakeholders
- GS9.** plan and organize the product lines
- GS10.** fix the appropriate price.
- GS11.** hire employees with the appropriate skill set and experience
- GS12.** predict the profit margin to be achieved by the business
- GS13.** decide target segment of the market
- GS14.** schedule production cycles for better efficiency of resources
- GS15.** planning of production efficiency based on manpower and equipment available
- GS16.** schedule market visits for surveys and feedback
- GS17.** gather information on preference and taste of the customer
- GS18.** interact with various types of customers and understand the trends
- GS19.** analyze and solve conflicts and problems of the business.
- GS20.** ensure that the problems do not arise repeatedly.
- GS21.** anticipate various problems/challenges that can crop up
- GS22.** analyze the market for increasing sales
- GS23.** spot errors and any other disruptions and communicate with solutions

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>People management</i>	3	6	-	-
PC1. arrange Interactive meetings of managers of sales and production teams and categorize the issues and feedbacks of both the teams	1	2	-	-
PC2. train the employees of his/her unit with the appropriate skills required to make market relevant and quality products	1	2	-	-
PC3. motivate the employees	1	2	-	-
<i>Product planning</i>	8	16	-	-
PC4. compile a report based on old production reports	1	2	-	-
PC5. address the issues faced in previous productions and try to resolve them	1	2	-	-
PC6. gather and analyze the cues from the market	1	2	-	-
PC7. ascertain the customer preference	1	2	-	-
PC8. develop product range lines based on current market preference	1	2	-	-
PC9. develop product range lines that are unique and able to price high	1	2	-	-
PC10. price the products according to market trends	1	2	-	-
PC11. identify the competent marketing strategy for the product range	1	2	-	-
<i>Procurement of raw materials</i>	7	14	-	-
PC12. list of the raw material s and prepare a B.O.M according to the product lines	1	2	-	-
PC13. ascertain the quantity and right price to procure the materials	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. identify the right locations/agents from where the raw materials can be procured	1	2	-	-
PC15. negotiate to get the best price	1	2	-	-
PC16. ensure quality materials are procured	1	2	-	-
PC17. ensure the procured materials are stored in appropriate conditions	1	2	-	-
PC18. compile a record of price quotations, POs, and bills of procurement for future reference	1	2	-	-
<i>Market interfacing</i>	7	14	-	-
PC19. maintain a healthy and professional relationship with vendor	1	2	-	-
PC20. the competitive market falls in order with the company policies of best price, quality, and delivery parameters	1	2	-	-
PC21. analyze the prevalent price for product lines	1	2	-	-
PC22. decide on the most effective means to access the market	1	2	-	-
PC23. plan for cost-effective transportation to the market	1	2	-	-
PC24. position the product according to market requirements	1	2	-	-
PC25. identify and address the expectations of customer	1	2	-	-
<i>Financial management</i>	4	8	-	-
PC26. analyze and ascertain the cost of production	1	2	-	-
PC27. maintain the book of accounts related to the business	1	2	-	-
PC28. maintain export documents like a letter of credit, custom clearance	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC29. identify cost-effective means of running the business	1	2	-	-
<i>Record keeping</i>	6	7	-	-
PC30. identify various aspects of business that require recording	1	2	-	-
PC31. design formats for recording	1	1	-	-
PC32. compile various records of all aspects of the business	1	1	-	-
PC33. maintain these records with periodic updations	1	1	-	-
PC34. maintain necessary documents as per local government and regulatory requirements	1	1	-	-
PC35. reframe the procurement strategy according to local scenarios like weather conditions, transport strikes, affected prices, etc.	1	1	-	-
NOS Total	35	65	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	HCS/N9941
NOS Name	Understanding Business Operations
Sector	Handicrafts and Carpet
Sub-Sector	
Occupation	Generic Handicrafts and Carpet
NSQF Level	2.5
Credits	1
Version	1.0
Last Reviewed Date	30/04/2025
Next Review Date	30/04/2028
NSQC Clearance Date	30/04/2025

Qualification Pack

DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

Qualification Pack

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services

Qualification Pack

- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Qualification Pack

HCS/N0602: Carry out the processes required to prepare the cast

Description

This OS unit is about preparing the ceramic cast using slip and permanent moulds forming technique for pottery and other ceramics, especially for shapes not easily made on a potter wheel.

Scope

The scope covers the following :

- Prepare the cast
- Casting the slip
- Achieve productivity and quality standards

Elements and Performance Criteria

Prepare the cast

To be competent, the user/individual on the job must be able to:

- PC1.** examine the drawing to check the design specifications and dimensions of the ceramic product
- PC2.** store the drawings in order to reduce the cause of damage by moisture, chemicals and fire
- PC3.** select the PPE required to handle the tools and equipment
- PC4.** place the permanent moulds at the work area with their openings upwards
- PC5.** determine the various casting parameters like temperature, pressure and air flow nozzles before casting the clay
- PC6.** collect the slip (semi-liquid clay) in slip bucket from the furnace in case there is no slip supply tank at the workplace

Casting the slip

To be competent, the user/individual on the job must be able to:

- PC7.** fill permanent moulds with slip using hose pipe from supply tank or container
- PC8.** pour excess slip from moulds when accumulation of clay around inside of mould opening indicates clay shell inside mould has built up to specified thickness
- PC9.** change position of moulds to produce uniform shell thickness of irregularly shaped items
- PC10.** place moulds upside down to drain after slip solidifies
- PC11.** operate the furnace for pre-heating process at uniform temperature
- PC12.** perform the casting process at desired temperature and pressure based on the raw materials and design specification
- PC13.** cool down the molten products at uniform temperature for further processing
- PC14.** open mould after trimming excess clay from inside mould opening
- PC15.** remove green casting from the mould when casting is firm enough to handle
- PC16.** examine the surfaces to remove mould marks using knives and wet sponge

Achieving productivity and quality standards

To be competent, the user/individual on the job must be able to:

Qualification Pack

- PC17.** analyze that the slip inside the mould has built up to specified thickness
- PC18.** remove casting from the mould only when it is firm enough to handle
- PC19.** examine the ceramic product based on the measurement, thickness, weight and surface finish
- PC20.** analyse any irregularity in the processes and take preventive steps whenever required
- PC21.** maintain records of inspection results and tests as per prescribed job sheet

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** company's policies on: human resource management, benefits and code of conduct
- KU2.** quality norms prescribed by the organization for mould making jobs
- KU3.** basic process followed for inspection of the pieces
- KU4.** basic principles of geometry and drawing
- KU5.** measuring instruments like Vernier callipers, micrometers and other measurements system
- KU6.** hazards and safety aspects involved in handling molten material
- KU7.** physical parameters like temperature, strength and shape etc.
- KU8.** different types of casting defects
- KU9.** importance of pouring the slip into mould at specified temperature and rate
- KU10.** usage of tools such as permanent moulds, slip bucket, knives and sponges
- KU11.** methods used for cutting, shearing, hammering, drilling which can repair pieces with minor defects

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** create small notes, work documents, diagrams for operators and helpers to help them understand the process
- GS2.** read equipment manuals and process documents
- GS3.** task lists, schedules, and work-loads with co-workers
- GS4.** support supervisor in using specific problem solving techniques and detailing out the problems
- GS5.** express the information, both technical and non-technical, to seniors and colleagues effectively.
- GS6.** identify and decide the critical tools and equipment to be used for casting
- GS7.** to prioritize and organise to achieve the daily targets
- GS8.** the customer preference, taste, etc. and accordingly make crafts
- GS9.** spot process disruptions and delays

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare the cast</i>	12	18	-	-
PC1. examine the drawing to check the design specifications and dimensions of the ceramic product	2	2	-	-
PC2. store the drawings in order to reduce the cause of damage by moisture, chemicals and fire	2	2	-	-
PC3. select the PPE required to handle the tools and equipment	2	3	-	-
PC4. place the permanent moulds at the work area with their openings upwards	2	3	-	-
PC5. determine the various casting parameters like temperature, pressure and air flow nozzles before casting the clay	2	4	-	-
PC6. collect the slip (semi-liquid clay) in slip bucket from the furnace in case there is no slip supply tank at the workplace	2	4	-	-
<i>Casting the slip</i>	17	31	-	-
PC7. fill permanent moulds with slip using hose pipe from supply tank or container	2	3	-	-
PC8. pour excess slip from moulds when accumulation of clay around inside of mould opening indicates clay shell inside mould has built up to specified thickness	2	3	-	-
PC9. change position of moulds to produce uniform shell thickness of irregularly shaped items	2	3	-	-
PC10. place moulds upside down to drain after slip solidifies	2	3	-	-
PC11. operate the furnace for pre-heating process at uniform temperature	2	3	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. perform the casting process at desired temperature and pressure based on the raw materials and design specification	2	3	-	-
PC13. cool down the molten products at uniform temperature for further processing	1	4	-	-
PC14. open mould after trimming excess clay from inside mould opening	1	3	-	-
PC15. remove green casting from the mould when casting is firm enough to handle	1	3	-	-
PC16. examine the surfaces to remove mould marks using knives and wet sponge	2	3	-	-
<i>Achieving productivity and quality standards</i>	8	14	-	-
PC17. analyze that the slip inside the mould has built up to specified thickness	2	3	-	-
PC18. remove casting from the mould only when it is firm enough to handle	2	3	-	-
PC19. examine the ceramic product based on the measurement, thickness, weight and surface finish	1	2	-	-
PC20. analyse any irregularity in the processes and take preventive steps whenever required	1	3	-	-
PC21. maintain records of inspection results and tests as per prescribed job sheet	2	3	-	-
NOS Total	37	63	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	HCS/N0602
NOS Name	Carry out the processes required to prepare the cast
Sector	Handicrafts and Carpet
Sub-Sector	Ceramics
Occupation	Casting
NSQF Level	2.5
Credits	2
Version	1.0
Last Reviewed Date	30/04/2025
Next Review Date	30/04/2028
NSQC Clearance Date	30/04/2025

Qualification Pack

HCS/N0816: Mix the ingredients to obtain the desired colours

Description

This OS unit is about mixing various oxides and other raw materials to obtain colouring mixture of ceramic products.

Scope

The scope covers the following :

- Arrange the required raw materials and ingredients to obtain the desired colour solution
- Perform colour mixing operation as per the standard procedures
- Check colour quality and colour productivity standards

Elements and Performance Criteria

Arrange the required raw materials and ingredients to obtain the desired colour solution

To be competent, the user/individual on the job must be able to:

- PC1.** list the raw materials and ingredients required for preparing the colour solution
- PC2.** arrange tools and equipment for processing the colour mixing operation depending upon the ball mix technique or manual technique
- PC3.** examine the job sheet based on the colour composition properties such as pre heating and post heating operations, dry and wet colour and oxides properties
- PC4.** collect the required raw materials and ingredients from the store as per the prescribed quantities and labelling
- PC5.** place all the collected raw materials and ingredients at the work place
- PC6.** deliver the remaining raw material and ingredients at designated location as per the SOP with proper labelling that ensures traceability and identification

Perform the colour mixing operation as per the standard procedures

To be competent, the user/individual on the job must be able to:

- PC7.** select the PPE required to handle the tools and equipment
- PC8.** monitor the functioning of ball mill based on the time duration and pressure if thickness of the colour is more dense
- PC9.** mix the raw materials and ingredients in the container of prescribed volume and size for manual mixing technique
- PC10.** perform the colour mixing operation as per the SOP to ensure appropriate colour formation is being achieved
- PC11.** examine the prepared colour composition based on color viscosity, oxides granules, nature of hue, and color saturation range
- PC12.** maintain and examine the overall functioning of the ball mill as per the standard operating procedures
- PC13.** turn off the ball mill at the right interval of time
- PC14.** remove the prepared colour formulation from the ball mill or from the container
- PC15.** clean tools and equipment after their use to prevent colour spots, stains and rust

Qualification Pack

Check colour quality and productivity standards

To be competent, the user/individual on the job must be able to:

- PC16.** apply the prepared colour solution on the raw piece to check the colour quality and its properties
- PC17.** compare the colour as per the colour quality standards during pre-heating or post heating operations
- PC18.** maintain records of inspection results and tests as per prescribed job sheet
- PC19.** review the periodical targets set by the supervisor

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** departmental standards, policies, and procedures followed in the company
- KU2.** basic chemical properties of various raw materials used
- KU3.** appropriate mixing techniques
- KU4.** basic calculation methods required for calculating raw materials and ingredients
- KU5.** the appropriate safety measures while handling the raw materials, tools and equipment
- KU6.** the working procedures of ball mill
- KU7.** explain the importance of maintaining records for traceability
- KU8.** units of measurement and temperatures
- KU9.** labelling and storing the colour mixes
- KU10.** risk and impact of not following defined procedures/work instructions
- KU11.** implications of delays in the process

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret symbols, readings, manuals, health and safety instructions, memos, reports and job cards
- GS2.** listen attentively to understand the information/ instructions being given
- GS3.** communicate about any delay (e.g. supplies, resolving issues) to the superior/customer
- GS4.** communicate the gaps in process, if any to supervisor
- GS5.** take the right raw materials according to the end product
- GS6.** detect problems in day to day tasks
- GS7.** plan and prioritise tasks to ensure timely completion
- GS8.** interpret the customer preference, demand, etc. and accordingly make job sheet
- GS9.** identify possible disruptions to work and take preventive measures

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Arrange the required raw materials and ingredients to obtain the desired colour solution</i>	11	18	-	-
PC1. list the raw materials and ingredients required for preparing the colour solution	2	3	-	-
PC2. arrange tools and equipment for processing the colour mixing operation depending upon the ball mix technique or manual technique	2	3	-	-
PC3. examine the job sheet based on the colour composition properties such as pre heating and post heating operations, dry and wet colour and oxides properties	2	3	-	-
PC4. collect the required raw materials and ingredients from the store as per the prescribed quantities and labelling	2	3	-	-
PC5. place all the collected raw materials and ingredients at the work place	2	3	-	-
PC6. deliver the remaining raw material and ingredients at designated location as per the SOP with proper labelling that ensures traceability and identification	1	3	-	-
<i>Perform the colour mixing operation as per the standard procedures</i>	17	34	-	-
PC7. select the PPE required to handle the tools and equipment	2	3	-	-
PC8. monitor the functioning of ball mill based on the time duration and pressure if thickness of the colour is more dense	2	3	-	-
PC9. mix the raw materials and ingredients in the container of prescribed volume and size for manual mixing technique	2	4	-	-
PC10. perform the colour mixing operation as per the SOP to ensure appropriate colour formation is being achieved	2	4	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. examine the prepared colour composition based on color viscosity, oxides granules, nature of hue, and color saturation range	2	4	-	-
PC12. maintain and examine the overall functioning of the ball mill as per the standard operating procedures	2	4	-	-
PC13. turn off the ball mill at the right interval of time	2	4	-	-
PC14. remove the prepared colour formulation from the ball mill or from the container	1	4	-	-
PC15. clean tools and equipment after their use to prevent colour spots, stains and rust	2	4	-	-
<i>Check colour quality and productivity standards</i>	4	16	-	-
PC16. apply the prepared colour solution on the raw piece to check the colour quality and its properties	1	4	-	-
PC17. compare the colour as per the colour quality standards during pre-heating or post heating operations	1	4	-	-
PC18. maintain records of inspection results and tests as per prescribed job sheet	1	4	-	-
PC19. review the periodical targets set by the supervisor	1	4	-	-
NOS Total	32	68	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	HCS/N0816
NOS Name	Mix the ingredients to obtain the desired colours
Sector	Handicrafts and Carpet
Sub-Sector	Ceramics
Occupation	Finishing and Painting
NSQF Level	2.5
Credits	2
Version	1.0
Last Reviewed Date	30/04/2025
Next Review Date	30/04/2028
NSQC Clearance Date	30/04/2025

Qualification Pack

HCS/N0703: Engobing of ceramics products

Description

This OS unit is about engobing the ceramic/blue pottery products prior to the sketching and colouring and prior to the firing process.

Scope

The scope covers the following :

- prepare astar/engobing mixture
- proper application of astar/engobing mixture
- ensure quality and productivity standards

Elements and Performance Criteria

Prepare astar/engobing mixture

To be competent, the user/individual on the job must be able to:

- PC1.** identify the raw materials for the astar/engobing mixture.
- PC2.** take the correct proportion of the raw materials to make astar/engobing mixture.
- PC3.** mix the materials appropriately to form the engobing mixture.
- PC4.** mix appropriate quantity of water to make the mixture the right liquid coating consistency.
- PC5.** set aside the engobing mixture carefully for further usage.
- PC6.** return the raw materials to the appropriate storage locations.

Proper application of astar/engobing mixture

To be competent, the user/individual on the job must be able to:

- PC7.** carefully take the ceramics/blue pottery products that have to be engobed.
- PC8.** engobe the products with the mixture.
- PC9.** ensure that no excess engobing mixture hangs-on to the product.
- PC10.** simultaneously twirl/move the product to ensure uniform application of the engobing mixture.
- PC11.** check the engobed products if the mixture has been applied uniformly and in right thickness.
- PC12.** set aside the engobed products carefully for drying.
- PC13.** clean the area after the engobing.

Ensure quality and productivity standards

To be competent, the user/individual on the job must be able to:

- PC14.** able to visually examine the raw materials for purity.
- PC15.** engobe the products in line with the production schedules.
- PC16.** ensure that there is no wastage of the materials.
- PC17.** able to visually examine the engobed product prior to the sketching and colouring.
- PC18.** achieve periodical targets set by the supervisor.
- PC19.** ensure there are no process delays.

Qualification Pack

PC20. ensure all time availability of materials and coordinate with stores for adequate inventory.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** company's policies on: personnel management, relevant legislation, standards, policies, and procedures followed in the company
- KU2.** organization's history and culture
- KU3.** organization structure
- KU4.** company's policies related to dress code and etiquette
- KU5.** the locations where the materials are typically stored
- KU6.** the key product lines of the company
- KU7.** basic chemical properties of various raw materials used
- KU8.** basic proportions of materials that can be used to make the mixture
- KU9.** the general way to mix the materials effectively
- KU10.** the recommended storage methodology for the raw materials
- KU11.** handling the raw materials as per material handling procedure
- KU12.** ensuring that there is no mal handling/accidents due to improper handling of the raw materials
- KU13.** storing the raw materials appropriately
- KU14.** managing his/her time to engobe all the products for a production schedule
- KU15.** appropriately scheduling the preparation of the engobing mixture so that when the time for application comes the mixture is ready
- KU16.** the recommended storage of the prepared engobing mixture
- KU17.** the science behind how the engobing mixture works
- KU18.** the necessity of engobing the ceramic/blue pottery products prior to the sketching and coloring
- KU19.** the techniques to ensure there is uniform astar application
- KU20.** the techniques to ensure that there is no excess astar in the product
- KU21.** the means to protect the astar from splattering
- KU22.** quality and safety standards to be followed.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** write basic process notes.
- GS2.** read about various engobing related information.
- GS3.** interact with employees to work efficiently.
- GS4.** interact with the supervisor.
- GS5.** communicate the gaps in process, if any to relevant colleagues.

Qualification Pack

- GS6.** take the right raw materials.
- GS7.** decide on the proportion.
- GS8.** decide on the right proportion of water.
- GS9.** ensure that there is uniform and smooth astar application.
- GS10.** to prioritize and organise to achieve the daily targets.
- GS11.** the customer preference, taste, etc and accordingly make crafts.
- GS12.** to solve work related problems.
- GS13.** how to analyse the material requirement.
- GS14.** spot process disruptions and delays.
- GS15.** report on any concerns to superiors without delay.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare astar/engobing mixture</i>	9	20	-	-
PC1. identify the raw materials for the astar/engobing mixture.	2	4	-	-
PC2. take the correct proportion of the raw materials to make astar/engobing mixture.	2	3	-	-
PC3. mix the materials appropriately to form the engobing mixture.	1	5	-	-
PC4. mix appropriate quantity of water to make the mixture the right liquid coating consistency.	2	4	-	-
PC5. set aside the engobing mixture carefully for further usage.	1	2	-	-
PC6. return the raw materials to the appropriate storage locations.	1	2	-	-
<i>Proper application of astar/engobing mixture</i>	12	29	-	-
PC7. carefully take the ceramics/blue pottery products that have to be engobed.	1	3	-	-
PC8. engobe the products with the mixture.	2	4	-	-
PC9. ensure that no excess engobing mixture hangs-on to the product.	2	5	-	-
PC10. simultaneously twirl/move the product to ensure uniform application of the engobing mixture.	2	5	-	-
PC11. check the engobed products if the mixture has been applied uniformly and in right thickness.	2	5	-	-
PC12. set aside the engobed products carefully for drying.	2	4	-	-
PC13. clean the area after the engobing.	1	3	-	-
<i>Ensure quality and productivity standards</i>	9	21	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. able to visually examine the raw materials for purity.	1	3	-	-
PC15. engobe the products in line with the production schedules.	1	3	-	-
PC16. ensure that there is no wastage of the materials.	1	3	-	-
PC17. able to visually examine the engobed product prior to the sketching and colouring.	2	4	-	-
PC18. achieve periodical targets set by the supervisor.	2	3	-	-
PC19. ensure there are no process delays.	1	3	-	-
PC20. ensure all time availability of materials and coordinate with stores for adequate inventory.	1	2	-	-
NOS Total	30	70	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	HCS/N0703
NOS Name	Engobing of ceramics products
Sector	Handicrafts and Carpet
Sub-Sector	Handicrafts (Ceramics)
Occupation	Blue Pottery Making
NSQF Level	2.5
Credits	2
Version	1.0
Last Reviewed Date	30/04/2025
Next Review Date	30/04/2028
NSQC Clearance Date	30/04/2025

Qualification Pack

HCS/N2104: Carry out furnace operation

Description

This unit is about combustion control inside the furnace for processing different materials as per product specifications. It also involves temperature control at the desired level, control of air compressor, and secondary air and level control inside the furnace.

Scope

The scope covers the following :

- control the temperature distribution inside the furnace
- control the secondary air and exhaust gas recovery
- control the flame length
- understand the furnace insulation
- achieve productivity, quality and safety standards

Elements and Performance Criteria

Control the temperature distribution inside the furnace

To be competent, the user/individual on the job must be able to:

- PC1.** operate the furnace with understanding of controls and specifications
- PC2.** understand the temperature requirement as per the product requirements and provide specifications
- PC3.** maintain temperature distribution inside the furnace at the optimum level
- PC4.** monitor the batch movement inside the furnace
- PC5.** control the cooling air inside the furnace
- PC6.** control the fuel at each port through burners
- PC7.** operate the air compressor

Control the secondary air and exhaust gas recovery

To be competent, the user/individual on the job must be able to:

- PC8.** control the preheating of secondary gas
- PC9.** monitor and control the flow of exhaust gas
- PC10.** understand and follow the procedure for exhaust gas recover

Control the flame length

To be competent, the user/individual on the job must be able to:

- PC11.** adjust burner capacity
- PC12.** control the volume of primary air
- PC13.** control the flame rotary angle as per requirement
- PC14.** achieve the required flame atomising pressure
- PC15.** adjust and control the secondary air

Understand the furnace insulation

Qualification Pack

To be competent, the user/individual on the job must be able to:

- PC16.** understand the insulation for the melting chamber
- PC17.** understand crown insulation
- PC18.** understand side wall insulation
- PC19.** prevent cold air from entering through inlet opening
- PC20.** ensure that outlet opening is closed during operation

Achieve productivity, quality and safety standards

To be competent, the user/individual on the job must be able to:

- PC21.** operate the furnace and ensure periodical targets are met as per supervisor instructions
- PC22.** deduce any maintenance requirement and ensure that the furnace is operational all the time
- PC23.** follow the standard operating procedures during furnace operation and avoid any damage
- PC24.** follow standard safety guidelines and mention dos and don'ts for other employees who comes near furnace operation area
- PC25.** ensure adequate fuel supply by informing procurement assistant to avoid work delay

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the company's policies
- KU2.** the company's code of conduct
- KU3.** organisational culture and typical product profile
- KU4.** company's reporting structure
- KU5.** the importance of individual's role in the workflow
- KU6.** company's daily/monthly targets for different product items
- KU7.** quality and safety standards
- KU8.** about the different types of melting furnaces
- KU9.** furnace operation and control
- KU10.** specifications of a furnace
- KU11.** basic upkeep of a furnace
- KU12.** temperature and control requirements for different products
- KU13.** combustion control inside furnace
- KU14.** control of primary and secondary air in the furnace
- KU15.** How to operate, control and maintain the furnace
- KU16.** batch movement inside furnace
- KU17.** temperature distribution inside furnace
- KU18.** furnace insulation
- KU19.** quality standards
- KU20.** safety regulations to be followed

Generic Skills (GS)

Qualification Pack

User/individual on the job needs to know how to:

- GS1.** measure furnace/flame temperature
- GS2.** measure air compressor
- GS3.** keep log book for the furnace
- GS4.** read and write the job requirement
- GS5.** read company policy documents
- GS6.** seek inputs at assessing problems
- GS7.** communicate in local language
- GS8.** educate and inform production team regarding temperature requirement for production of different kinds of products
- GS9.** educate on precautions to be taken during production to ensure safe working surroundings
- GS10.** give feedback to production manager regarding the quality of products
- GS11.** decide on material requirement for furnace operation
- GS12.** plan for material and time requirement for furnace operation
- GS13.** meet customer's preferences on craft products
- GS14.** identify any problems/issues in the furnace and find solution
- GS15.** anticipate problems and take necessary actions
- GS16.** seek assistance from supervisors and technicians to resolve problems
- GS17.** analyse the temperature requirement for various products
- GS18.** spot process disruptions and delays

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Control the temperature distribution inside the furnace</i>	7	21	-	-
PC1. operate the furnace with understanding of controls and specifications	1	3	-	-
PC2. understand the temperature requirement as per the product requirements and provide specifications	1	3	-	-
PC3. maintain temperature distribution inside the furnace at the optimum level	1	3	-	-
PC4. monitor the batch movement inside the furnace	1	3	-	-
PC5. control the cooling air inside the furnace	1	3	-	-
PC6. control the fuel at each port through burners	1	3	-	-
PC7. operate the air compressor	1	3	-	-
<i>Control the secondary air and exhaust gas recovery</i>	3	9	-	-
PC8. control the preheating of secondary gas	1	3	-	-
PC9. monitor and control the flow of exhaust gas	1	3	-	-
PC10. understand and follow the procedure for exhaust gas recover	1	3	-	-
<i>Control the flame length</i>	5	15	-	-
PC11. adjust burner capacity	1	3	-	-
PC12. control the volume of primary air	1	3	-	-
PC13. control the flame rotary angle as per requirement	1	3	-	-
PC14. achieve the required flame atomising pressure	1	3	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC15. adjust and control the secondary air	1	3	-	-
<i>Understand the furnace insulation</i>	5	15	-	-
PC16. understand the insulation for the melting chamber	1	3	-	-
PC17. understand crown insulation	1	3	-	-
PC18. understand side wall insulation	1	3	-	-
PC19. prevent cold air from entering through inlet opening	1	3	-	-
PC20. ensure that outlet opening is closed during operation	1	3	-	-
<i>Achieve productivity, quality and safety standards</i>	5	15	-	-
PC21. operate the furnace and ensure periodical targets are met as per supervisor instructions	1	3	-	-
PC22. deduce any maintenance requirement and ensure that the furnace is operational all the time	1	3	-	-
PC23. follow the standard operating procedures during furnace operation and avoid any damage	1	3	-	-
PC24. follow standard safety guidelines and mention dos and don'ts for other employees who comes near furnace operation area	1	3	-	-
PC25. ensure adequate fuel supply by informing procurement assistant to avoid work delay	1	3	-	-
NOS Total	25	75	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	HCS/N2104
NOS Name	Carry out furnace operation
Sector	Handicrafts and Carpet
Sub-Sector	Ceramics, Glassware
Occupation	Furnace and Oven Operation-Glassware
NSQF Level	2.5
Credits	2
Version	1.0
Last Reviewed Date	30/04/2025
Next Review Date	30/04/2028
NSQC Clearance Date	30/04/2025

Qualification Pack

HCS/N0817: Perform Glazing of Ceramics Products

Description

This OS unit is about glazing the ceramic products after the sketching and colouring and prior to the firing process.

Scope

The scope covers the following :

- prepare the glaze mixture
- proper application of glaze mixture
- ensure quality and productivity standards

Elements and Performance Criteria

prepare the glaze mixture

To be competent, the user/individual on the job must be able to:

- PC1.** identify the raw materials for the glaze mixture according to the type of glaze required
- PC2.** take the correct proportion of the raw materials to make the glaze mixture.
- PC3.** treat the raw materials in the appropriate manner prior to making the mixture
- PC4.** mix the materials appropriately to form the glaze mixture
- PC5.** set aside the glaze mixture carefully for further usage
- PC6.** return the raw materials to the appropriate storage locations

proper application of glaze mixture

To be competent, the user/individual on the job must be able to:

- PC7.** carefully take the ceramics that have to be dipped
- PC8.** dip the products in the glaze mixture
- PC9.** ensure that no excess glaze mixture hangs-on to the product
- PC10.** simultaneously twirl/move the product to ensure uniform application of the glaze mixture
- PC11.** check the glazed products if the mixture has been applied uniformly and in right thickness
- PC12.** set aside the glazed products carefully for drying
- PC13.** clean the area after the glazing

ensure quality and productivity standards

To be competent, the user/individual on the job must be able to:

- PC14.** able to visually examine the raw materials for purity
- PC15.** glaze the products in line with the production schedules
- PC16.** ensure that there is no wastage of the materials
- PC17.** able to visually examine the glazed product prior to the firing / drying
- PC18.** achieve periodical targets set by the supervisor
- PC19.** ensure there are no process delays
- PC20.** ensure all time availability of materials and coordinate with stores for adequate inventory

Qualification Pack

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** company's policies on: personnel management, relevant legislation, standards, policies, and procedures followed in the company
- KU2.** organizations history and culture
- KU3.** organisation structure
- KU4.** company's policies related to dress code and etiquette
- KU5.** the locations where the materials are typically stored
- KU6.** the key product lines of the company
- KU7.** basic chemical properties of various raw materials used
- KU8.** basic proportions of materials that can be used to make the mixture
- KU9.** the general way to mix the materials effectively
- KU10.** the recommended storage methodology for the raw materials
- KU11.** handling the raw materials appropriately
- KU12.** ensuring that there is no mal handling/accidents due to improper handling of the raw materials.
- KU13.** storing the raw materials appropriately
- KU14.** managing his/her time to glaze all the products for a production schedule
- KU15.** appropriately scheduling the preparation of the glazing mixture so that when the time for the application comes the mixture is ready
- KU16.** the recommended storage of the prepared glaze mixture
- KU17.** the science behind how the glaze mixture works along with the ceramic product to give the sheen post the firing
- KU18.** the techniques to ensure there is uniform glaze application
- KU19.** the techniques to ensure that there is no excess glaze in the product
- KU20.** the means to protect the glaze from splattering
- KU21.** material handling procedure
- KU22.** ceramic craft processes
- KU23.** quality and safety standards associated with the glazing process

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** write basic process notes
- GS2.** read about various glazing related information
- GS3.** interact with employees to work efficiently
- GS4.** interact with the supervisor
- GS5.** communicate the gaps in the process, if any to relevant colleagues
- GS6.** take the right raw materials according to the glaze

Qualification Pack

- GS7.** decide on the proportion
- GS8.** ensure that there is uniform and smooth glaze application
- GS9.** to prioritize and organise to achieve the daily targets
- GS10.** the customer preference, taste, etc and accordingly make crafts
- GS11.** to solve work related problems
- GS12.** how to analyse the material requirement
- GS13.** spot process disruptions and delays
- GS14.** report on any concerns to superiors without delay

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>prepare the glaze mixture</i>	6	24	-	-
PC1. identify the raw materials for the glaze mixture according to the type of glaze required	1	4	-	-
PC2. take the correct proportion of the raw materials to make the glaze mixture.	1	4	-	-
PC3. treat the raw materials in the appropriate manner prior to making the mixture	1	4	-	-
PC4. mix the materials appropriately to form the glaze mixture	1	4	-	-
PC5. set aside the glaze mixture carefully for further usage	1	4	-	-
PC6. return the raw materials to the appropriate storage locations	1	4	-	-
<i>proper application of glaze mixture</i>	7	28	-	-
PC7. carefully take the ceramics that have to be dipped	1	4	-	-
PC8. dip the products in the glaze mixture	1	4	-	-
PC9. ensure that no excess glaze mixture hangs-on to the product	1	4	-	-
PC10. simultaneously twirl/move the product to ensure uniform application of the glaze mixture	1	4	-	-
PC11. check the glazed products if the mixture has been applied uniformly and in right thickness	1	4	-	-
PC12. set aside the glazed products carefully for drying	1	4	-	-
PC13. clean the area after the glazing	1	4	-	-
<i>ensure quality and productivity standards</i>	7	28	-	-
PC14. able to visually examine the raw materials for purity	1	4	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC15. glaze the products in line with the production schedules	1	4	-	-
PC16. ensure that there is no wastage of the materials	1	4	-	-
PC17. able to visually examine the glazed product prior to the firing / drying	1	4	-	-
PC18. achieve periodical targets set by the supervisor	1	4	-	-
PC19. ensure there are no process delays	1	4	-	-
PC20. ensure all time availability of materials and coordinate with stores for adequate inventory	1	4	-	-
NOS Total	20	80	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	HCS/N0817
NOS Name	Perform Glazing of Ceramics Products
Sector	Handicrafts and Carpet
Sub-Sector	Ceramics
Occupation	Finishing and Painting
NSQF Level	2.5
Credits	2
Version	1.0
Last Reviewed Date	30/04/2025
Next Review Date	30/04/2028
NSQC Clearance Date	30/04/2025

Qualification Pack

HCS/N0502: Perform jiggering of Dough

Description

This OS unit is about jiggering of ceramic dough using jigger equipment to give shape as per the prescribed standards.

Scope

The scope covers the following :

- prepare jigger machine
- perform jiggering of dough as per the standards
- check quality and productivity standards

Elements and Performance Criteria

Prepare the jigger machine

To be competent, the user/individual on the job must be able to:

- PC1.** clean jigger machine before jiggering
- PC2.** ensure that all previously used calibrations are removed to prepare for jiggering
- PC3.** analyze the dimensions of the product that has to be shaped by jigger machine
- PC4.** collect the appropriate amount of clay mass from the pug mill
- PC5.** identify the appropriate position to operate jigger machine

Perform jiggering of dough as per the standard procedures

To be competent, the user/individual on the job must be able to:

- PC6.** ensure use of PPE to avoid any debris affects
- PC7.** perform the step of calibrating jigger machine as per the shape/dimension required
- PC8.** Identify and select appropriate enablers like hat block to obtain specification
- PC9.** fix enabler to the jigger machine
- PC10.** load dough in appropriate quantities in jigger machine as per the final product requirement
- PC11.** operate jigger machine as per standards to shape the dough
- PC12.** ensure to use water in prescribed quantities if required in the shaping of dough
- PC13.** perform the step of touching the shaping dough to ensure that the desired shape is being achieved
- PC14.** ensure that the shape is as per specified requirement
- PC15.** explain the importance of turning off the jigger machine at the right interval of time
- PC16.** remove lumps of excess dough from jigger
- PC17.** carefully place the shaped product at storage locations for inspection
- PC18.** maintain and examine the overall functioning of the jigger machine as per the standard

Check quality and productivity standards

To be competent, the user/individual on the job must be able to:

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- PC19.** demonstrate the inspection process based on the quality standard parameters to ensure appropriate shape is being achieved
- PC20.** demonstrate the steps to be followed while maintaining records of inspection results/tests
- PC21.** ensure that there is no wastage of materials
- PC22.** prepare a sample template to collect information during inspection based on the final shape
- PC23.** achieve periodical targets set by the supervisor
- PC24.** perform sample test from the prepared shape to check the quality
- PC25.** identify different parameters to ensure the appropriate amount of inventory is being stored at the time of production

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** personnel management, relevant legislation, standards, policies, and procedures followed in the company
- KU2.** the locations where materials are typically stored
- KU3.** the key product lines of the company
- KU4.** appropriate shaping techniques
- KU5.** basic calculation methods required for calculating clay requirement
- KU6.** the appropriate safety measures while handling the raw materials
- KU7.** handle the use of tools and equipment
- KU8.** ensuring that there is no mal handling/accidents due to improper handling of the raw materials
- KU9.** handling the equipment like jigger machine appropriately
- KU10.** the recommended storage of the prepared shape
- KU11.** basic up keep of jigger machine
- KU12.** risk and impact of not following defined procedures/work instructions
- KU13.** the hierarchy for reporting identified problems
- KU14.** implications of delays in the process

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret symbols, readings, manuals, health and safety instructions, memos, reports and job cards
- GS2.** listen attentively to understand the information/ instructions being given
- GS3.** communicate about any delay (e.g. supplies, resolving issues) to the superior/customer
- GS4.** communicate the gaps in process, if any to supervisor
- GS5.** take the right raw materials according to the end product
- GS6.** detect problems in day to day tasks
- GS7.** plan and prioritise tasks to ensure timely completion

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GS8. interpret the customer preference, demand, etc. and accordingly make job sheet

GS9. identify possible disruptions to work and take preventive measures

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare the jigger machine</i>	5	15	-	-
PC1. clean jigger machine before jiggering	1	3	-	-
PC2. ensure that all previously used calibrations are removed to prepare for jiggering	1	3	-	-
PC3. analyze the dimensions of the product that has to be shaped by jigger machine	1	3	-	-
PC4. collect the appropriate amount of clay mass from the pug mill	1	3	-	-
PC5. identify the appropriate position to operate jigger machine	1	3	-	-
<i>Perform jiggering of dough as per the standard procedures</i>	13	39	-	-
PC6. ensure use of PPE to avoid any debris affects	1	3	-	-
PC7. perform the step of calibrating jigger machine as per the shape/dimension required	1	3	-	-
PC8. Identify and select appropriate enablers like hat block to obtain specification	1	3	-	-
PC9. fix enabler to the jigger machine	1	3	-	-
PC10. load dough in appropriate quantities in jigger machine as per the final product requirement	1	3	-	-
PC11. operate jigger machine as per standards to shape the dough	1	3	-	-
PC12. ensure to use water in prescribed quantities if required in the shaping of dough	1	3	-	-
PC13. perform the step of touching the shaping dough to ensure that the desired shape is being achieved	1	3	-	-
PC14. ensure that the shape is as per specified requirement	1	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC15. explain the importance of turning off the jigger machine at the right interval of time	1	3	-	-
PC16. remove lumps of excess dough from jigger	1	3	-	-
PC17. carefully place the shaped product at storage locations for inspection	1	3	-	-
PC18. maintain and examine the overall functioning of the jigger machine as per the standard	1	3	-	-
<i>Check quality and productivity standards</i>	7	21	-	-
PC19. demonstrate the inspection process based on the quality standard parameters to ensure appropriate shape is being achieved	1	3	-	-
PC20. demonstrate the steps to be followed while maintaining records of inspection results/tests	1	3	-	-
PC21. ensure that there is no wastage of materials	1	3	-	-
PC22. prepare a sample template to collect information during inspection based on the final shape	1	3	-	-
PC23. achieve periodical targets set by the supervisor	1	3	-	-
PC24. perform sample test from the prepared shape to check the quality	1	3	-	-
PC25. identify different parameters to ensure the appropriate amount of inventory is being stored at the time of production	1	3	-	-
NOS Total	25	75	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	HCS/N0502
NOS Name	Perform jiggering of Dough
Sector	Handicrafts and Carpet
Sub-Sector	Handicrafts (Ceramics)
Occupation	Jiggering
NSQF Level	2.5
Credits	2
Version	1.0
Last Reviewed Date	30/04/2025
Next Review Date	30/04/2028
NSQC Clearance Date	30/04/2025

Qualification Pack

HCS/N0408: Prepare the Mould for production

Description

This OS unit is about preparing required raw materials, making mould with the help of model and canister for production line.

Scope

The scope covers the following :

- prepare required raw materials for mould making
- perform the standard process of mould making
- perform the standard process of finishing
- check mould quality and productivity standards

Elements and Performance Criteria

Prepare the moulding mix

To be competent, the user/individual on the job must be able to:

- PC1.** locate the raw materials like plaster of paris, water, etc
- PC2.** select the appropriate materials for making moulds
- PC3.** ascertain the right proportion of the raw materials
- PC4.** mix these raw materials as per the formulation / standard procedures
- PC5.** ensure that the mix is of adequate consistency
- PC6.** ensure that the mix does not harden prior to the mould making
- PC7.** store the moulding mix in locations that can be easily accessed for moulding
- PC8.** return the excess raw materials to the store for further usage
- PC9.** clean the area after mixing

Moulding based on model

To be competent, the user/individual on the job must be able to:

- PC10.** receive the model for moulding
- PC11.** analyse and study the model before moulding
- PC12.** ascertain the model specifications before readying for moulding
- PC13.** ready the canisters for placing the model
- PC14.** place the model inside the canisters carefully
- PC15.** access the moulding mix carefully
- PC16.** take the appropriate amount of the mix for moulding
- PC17.** pour the mix in a uniform manner in the canisters with the model
- PC18.** smoothen the surface of the canister for uniform spread of the mix
- PC19.** ensure that the leakage of the excess mix is halted
- PC20.** set aside the canister for an appropriate time for the setting
- PC21.** monitor the time to ensure proper setting

Qualification Pack

Finishing the mould

To be competent, the user/individual on the job must be able to:

- PC22.** remove the mould from the canister after the requisite time has elapsed
- PC23.** ensure that mould does not suffer damage while removing
- PC24.** clean the mould by smoothing the overall circumference
- PC25.** wipe the mould
- PC26.** clean the canisters
- PC27.** return the canisters to the designated store area
- PC28.** clean the overall area
- PC29.** able to visually examine the moulds for conformity with the product
- PC30.** ensure that there is no wastage of the materials
- PC31.** able to visually examine if the final mould looks neat

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** company's policies on: personnel management, relevant legislation, standards, policies, and procedures followed in the company
- KU2.** organizations history and culture
- KU3.** organisation structure
- KU4.** company's policies related to dress code and etiquette
- KU5.** the locations where the materials are typically stored
- KU6.** the key product lines of the company
- KU7.** mixing techniques
- KU8.** the properties of the raw materials
- KU9.** the significance of the moulding process in the production
- KU10.** the art of working neatly
- KU11.** the right quantity of mixes that are typically required to make the moulds
- KU12.** the typical setting duration and time
- KU13.** finishing skills
- KU14.** examining techniques
- KU15.** appropriate cleaning techniques
- KU16.** the ability to work optimally
- KU17.** the quantity of mould mixture required
- KU18.** canister handling skills

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read about various designing related information

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- GS2.** write basic process notes
- GS3.** interact with employees to work efficiently
- GS4.** interact with the supervisor
- GS5.** communicate the gaps in the process, if any to relevant colleagues
- GS6.** decide on the raw material's proportion
- GS7.** decide on the setting time
- GS8.** finalize the quantity of the mix
- GS9.** share the workload and achieve the targets
- GS10.** spot process disruptions and delays
- GS11.** report on any concerns to superiors without delay
- GS12.** reduce repetition of errors
- GS13.** improve work processes
- GS14.** manage his/her time to make all the requisite moulds
- GS15.** appropriately schedule mould mix making so that moulding can be scheduled appropriately

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare the moulding mix</i>	9	18	-	-
PC1. locate the raw materials like plaster of paris, water, etc	1	2	-	-
PC2. select the appropriate materials for making moulds	1	2	-	-
PC3. ascertain the right proportion of the raw materials	1	2	-	-
PC4. mix these raw materials as per the formulation / standard procedures	1	2	-	-
PC5. ensure that the mix is of adequate consistency	1	2	-	-
PC6. ensure that the mix does not harden prior to the mould making	1	2	-	-
PC7. store the moulding mix in locations that can be easily accessed for moulding	1	2	-	-
PC8. return the excess raw materials to the store for further usage	1	2	-	-
PC9. clean the area after mixing	1	2	-	-
<i>Moulding based on model</i>	12	24	-	-
PC10. receive the model for moulding	1	2	-	-
PC11. analyse and study the model before moulding	1	2	-	-
PC12. ascertain the model specifications before readying for moulding	1	2	-	-
PC13. ready the canisters for placing the model	1	2	-	-
PC14. place the model inside the canisters carefully	1	2	-	-
PC15. access the moulding mix carefully	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC16. take the appropriate amount of the mix for moulding	1	2	-	-
PC17. pour the mix in a uniform manner in the canisters with the model	1	2	-	-
PC18. smoothen the surface of the canister for uniform spread of the mix	1	2	-	-
PC19. ensure that the leakage of the excess mix is halted	1	2	-	-
PC20. set aside the canister for an appropriate time for the setting	1	2	-	-
PC21. monitor the time to ensure proper setting	1	2	-	-
<i>Finishing the mould</i>	10	27	-	-
PC22. remove the mould from the canister after the requisite time has elapsed	1	2	-	-
PC23. ensure that mould does not suffer damage while removing	1	2	-	-
PC24. clean the mould by smoothing the overall circumference	1	2	-	-
PC25. wipe the mould	1	3	-	-
PC26. clean the canisters	1	3	-	-
PC27. return the canisters to the designated store area	1	3	-	-
PC28. clean the overall area	1	3	-	-
PC29. able to visually examine the moulds for conformity with the product	1	3	-	-
PC30. ensure that there is no wastage of the materials	1	3	-	-
PC31. able to visually examine if the final mould looks neat	1	3	-	-
NOS Total	31	69	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	HCS/N0408
NOS Name	Prepare the Mould for production
Sector	Handicrafts and Carpet
Sub-Sector	Ceramics
Occupation	Moulding
NSQF Level	2.5
Credits	2
Version	1.0
Last Reviewed Date	30/04/2025
Next Review Date	30/04/2028
NSQC Clearance Date	30/04/2025

Qualification Pack

HCS/N0303: Perform mixing of raw materials using equipment

Description

This OS unit is about mixing various raw materials like ball clay, kaolin, and quartz, etc. to obtain the slip for subsequent phases of production using appropriate equipment.

Scope

The scope covers the following :

- prepare required raw materials
- perform raw material mixing operation as per the standard procedures
- check mixture quality as per productivity standards

Elements and Performance Criteria

Prepare required raw materials for mixing

To be competent, the user/individual on the job must be able to:

- PC1.** place raw materials like ball clay, water, kaolin, etc. for making the slip like quartz, feldspar, ball clay, etc
- PC2.** store the different types of raw materials at designated locations
- PC3.** collect raw materials in prescribed quantities like ball clay 25 %, Kaolin 28 %, and quartz 32%.
- PC4.** load raw materials into the ball mill to get the appropriate proportion of grinded mixture
- PC5.** deliver the remaining raw material to the designated storage locations

Perform mixing operation as per the standard procedures

To be competent, the user/individual on the job must be able to:

- PC6.** operate the ball mill to grind the loaded raw materials
- PC7.** perform grinding process
- PC8.** shift grinded mixture from the ball mill to the next stage of production to the blunger
- PC9.** calibrate the blunger appropriately to get the slurry as per prescribed standards
- PC10.** perform appropriate mixing of slurry in blunger
- PC11.** carefully remove the prepared slurry from blunger and shift it to filtering equipment
- PC12.** ensure that the filtering equipment is calibrated to absorb sufficient amount of water from the slurry to make the 'paste' for further production stages
- PC13.** collect the 'paste' from the filtering press to be fed into pug mill for further processing
- PC14.** calibrate the pug mill to absorb all unwanted moisture
- PC15.** load the paste into the pug mill in appropriate quantities
- PC16.** operate the pug mill to ensure the slip comes out of it
- PC17.** collect the slip from the pug mill and keep it aside for further processing
- PC18.** ensure maintenance of equipment and smooth operation
- PC19.** switch off the equipment at the appropriate time

Qualification Pack

PC20. perform appropriate cleaning procedures to clean the area after mixing

Check quality and productivity standards

To be competent, the user/individual on the job must be able to:

PC21. inspect the prepared mixture based on the quality standard parameters to ensure purity

PC22. ensure mixtures are loaded as per recommended procedure to ensure no unwanted spillage and slippage

PC23. demonstrate mixing of products in line with the production schedules

PC24. ensure that there is no wastage of the materials

PC25. ensure there are no process delays as mixing is the first step for further processing

PC26. inspect the entire process of mixing to see if it is adhering to the usual production schedule

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. personnel management, relevant legislation, standards, policies, and procedures followed in the company

KU2. company's policies related to dress code and etiquette

KU3. the locations where materials are typically stored

KU4. the key product lines of the company

KU5. basic chemical properties of various raw materials used

KU6. appropriate mixing techniques

KU7. basic calculation methods required for calculating raw materials

KU8. the appropriate safety measures while handling the raw materials

KU9. handle the use of sharp tools and equipment

KU10. ensuring that there is no mal handling/accidents due to improper handling of the raw materials

KU11. handling the equipment like ball mill, blunger, pug mill, and filter press appropriately

KU12. controls and basic up keeping of ball mill

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. write down observations if any

GS2. write down manual mixing techniques

GS3. read and interpret various colour mixing related information

GS4. read about various mixing related information

GS5. read about operating the ball mill, blunger, pug mill, and filter press

GS6. interact with employees to work efficiently

GS7. make decisions pertaining to the concerned area of workplace

GS8. communicate the gaps in process, if any to supervisor

Qualification Pack

- GS9.** take the right raw materials according to the end product
- GS10.** detect problems in day to day tasks
- GS11.** maintain the working schedule appropriately in order to achieve the daily targets
- GS12.** interpret the customer preference, demand, etc. and accordingly make job sheet
- GS13.** analyze and detect any potential problems which could arise during operation
- GS14.** analyze the material requirement
- GS15.** spot process disruptions and delays
- GS16.** follow instructions and work on areas of improvement identified

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare required raw materials for mixing</i>	5	11	-	-
PC1. place raw materials like ball clay, water, kaolin, etc. for making the slip like quartz, feldspar, ball clay, etc	1	1	-	-
PC2. store the different types of raw materials at designated locations	1	1	-	-
PC3. collect raw materials in prescribed quantities like ball clay 25 %, Kaolin 28 %, and quartz 32%.	1	3	-	-
PC4. load raw materials into the ball mill to get the appropriate proportion of grinded mixture	1	3	-	-
PC5. deliver the remaining raw material to the designated storage locations	1	3	-	-
<i>Perform mixing operation as per the standard procedures</i>	15	45	-	-
PC6. operate the ball mill to grind the loaded raw materials	1	3	-	-
PC7. perform grinding process	1	3	-	-
PC8. shift grinded mixture from the ball mill to the next stage of production to the blunger	1	3	-	-
PC9. calibrate the blunger appropriately to get the slurry as per prescribed standards	1	3	-	-
PC10. perform appropriate mixing of slurry in blunger	1	3	-	-
PC11. carefully remove the prepared slurry from blunger and shift it to filtering equipment	1	3	-	-
PC12. ensure that the filtering equipment is calibrated to absorb sufficient amount of water from the slurry to make the 'paste' for further production stages	1	3	-	-
PC13. collect the 'paste' from the filtering press to be fed into pug mill for further processing	1	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. calibrate the pug mill to absorb all unwanted moisture	1	3	-	-
PC15. load the paste into the pug mill in appropriate quantities	1	3	-	-
PC16. operate the pug mill to ensure the slip comes out of it	1	3	-	-
PC17. collect the slip from the pug mill and keep it aside for further processing	1	3	-	-
PC18. ensure maintenance of equipment and smooth operation	1	3	-	-
PC19. switch off the equipment at the appropriate time	1	3	-	-
PC20. perform appropriate cleaning procedures to clean the area after mixing	1	3	-	-
<i>Check quality and productivity standards</i>	6	18	-	-
PC21. inspect the prepared mixture based on the quality standard parameters to ensure purity	1	3	-	-
PC22. ensure mixtures are loaded as per recommended procedure to ensure no unwanted spillage and slippage	1	3	-	-
PC23. demonstrate mixing of products in line with the production schedules	1	3	-	-
PC24. ensure that there is no wastage of the materials	1	3	-	-
PC25. ensure there are no process delays as mixing is the first step for further processing	1	3	-	-
PC26. inspect the entire process of mixing to see if it is adhering to the usual production schedule	1	3	-	-
NOS Total	26	74	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	HCS/N0303
NOS Name	Perform mixing of raw materials using equipment
Sector	Handicrafts and Carpet
Sub-Sector	Ceramics
Occupation	Mixing and Milling
NSQF Level	2.5
Credits	1
Version	1.0
Last Reviewed Date	30/04/2025
Next Review Date	30/04/2028
NSQC Clearance Date	30/04/2025

Qualification Pack

HCS/N0304: Perform Manual mixing

Description

This OS unit is about mixing various raw materials like ball clay, kaolin, and quartz, etc. to obtain the slip for subsequent phases of production manually.

Scope

The scope covers the following :

- Prepare required raw materials for manual mixing
- Perform mixing operation as per the standard procedures
- Check mixture quality as per productivity standards

Elements and Performance Criteria

prepare required raw materials for manual mixing

To be competent, the user/individual on the job must be able to:

- PC1.** place the raw materials like ball clay, quartz powder, green or white glass powder, fuller's earth/multani mitti, katira/banyan tree gum, sazzi mitti/sodium sulphate, water for making the slip.
- PC2.** perform the standard organizational procedures to be followed to store different types of raw materials at designated locations
- PC3.** collect the raw materials in prescribed quantities like ball clay 25 %, Kaolin 28 %, and quartz 32%.
- PC4.** perform preparation of each ingredient as per standards for the next stage of mixing like crushing, etc.
- PC5.** carefully place the raw material in chakki/stone pulverizer for grinding process
- PC6.** deliver the remaining raw material to the designated storage locations

Perform mixing operation manually as per the standard procedures

To be competent, the user/individual on the job must be able to:

- PC7.** grind the raw materials using tools like chakki/stone pulverizer
- PC8.** ensure grinding of the raw materials is of right size to form slip
- PC9.** carefully add the appropriate quantity of water after the grinding to the raw materials
- PC10.** perform manual mixing of raw materials with hands along with water to form the slip
- PC11.** ensure that slip is malleable
- PC12.** perform appropriate cleaning procedures to clean the area after mixing

Check quality and productivity standards

To be competent, the user/individual on the job must be able to:

- PC13.** inspect quality standard parameters to ensure purity
- PC14.** ensure that mixing of products in line with the production schedules
- PC15.** ensure that there is no wastage of the materials
- PC16.** ensure there are no process delays as mixing is the first step for further processing

Qualification Pack

PC17. inspect the entire process of mixing to see if it is adhering to the usual production schedule

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** personnel management, relevant legislation, standards, policies, and procedures followed in the company
- KU2.** the locations where materials are typically stored
- KU3.** the key product lines of the company
- KU4.** basic chemical properties of various raw materials used
- KU5.** appropriate mixing techniques
- KU6.** basic calculation methods required for calculating raw materials
- KU7.** mixing material formula / description sheet
- KU8.** weighing of materials
- KU9.** appropriate safety measures while handling the raw materials
- KU10.** handle the use of sharp tools and equipment
- KU11.** ensuring that there is no mal handling/accidents due to improper handling of the raw materials
- KU12.** handling the equipment like ball mill, blunger, pug mill, and filter press appropriately
- KU13.** controls and basic up keeping of ball mill

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** note down manual mixing techniques
- GS2.** read and interpret various mixing related information
- GS3.** read about operating chakki/stone pulverizer
- GS4.** interact with employees to work efficiently
- GS5.** make decisions pertaining to the concerned area of workplace
- GS6.** communicate the gaps in process, if any to supervisor
- GS7.** take the right raw materials according to the end product
- GS8.** detect problems in day to day tasks
- GS9.** maintain the working schedule appropriately in order to achieve the daily targets
- GS10.** analyze and detect any potential problems which could arise during operation
- GS11.** analyze the material requirement
- GS12.** spot process disruptions and delays
- GS13.** follow instructions and work on areas of improvement identified

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>prepare required raw materials for manual mixing</i>	9	24	-	-
PC1. place the raw materials like ball clay, quartz powder, green or white glass powder, fuller's earth/multani mitti, katira/banyan tree gum, sazzi mitti/sodium sulphate, water for making the slip.	1	4	-	-
PC2. perform the standard organizational procedures to be followed to store different types of raw materials at designated locations	1	4	-	-
PC3. collect the raw materials in prescribed quantities like ball clay 25 %, Kaloin 28 %, and quartz 32%.	1	4	-	-
PC4. perform preparation of each ingredient as per standards for the next stage of mixing like crushing, etc.	2	4	-	-
PC5. carefully place the raw material in chakki/stone pulverizer for grinding process	2	4	-	-
PC6. deliver the remaining raw material to the designated storage locations	2	4	-	-
<i>Perform mixing operation manually as per the standard procedures</i>	12	24	-	-
PC7. grind the raw materials using tools like chakki/stone pulverizer	2	4	-	-
PC8. ensure grinding of the raw materials is of right size to form slip	2	4	-	-
PC9. carefully add the appropriate quantity of water after the grinding to the raw materials	2	4	-	-
PC10. perform manual mixing of raw materials with hands along with water to form the slip	2	4	-	-
PC11. ensure that slip is malleable	2	4	-	-
PC12. perform appropriate cleaning procedures to clean the area after mixing	2	4	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Check quality and productivity standards</i>	10	21	-	-
PC13. inspect quality standard parameters to ensure purity	2	4	-	-
PC14. ensure that mixing of products in line with the production schedules	2	4	-	-
PC15. ensure that there is no wastage of the materials	2	4	-	-
PC16. ensure there are no process delays as mixing is the first step for further processing	2	4	-	-
PC17. inspect the entire process of mixing to see if it is adhering to the usual production schedule	2	5	-	-
NOS Total	31	69	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	HCS/N0304
NOS Name	Perform Manual mixing
Sector	Handicrafts and Carpet
Sub-Sector	Ceramics
Occupation	Mixing and Milling
NSQF Level	2.5
Credits	1
Version	1.0
Last Reviewed Date	30/04/2025
Next Review Date	30/04/2028
NSQC Clearance Date	30/04/2025

Qualification Pack

HCS/N0109: Perform Slip Trailing

Description

This unit focuses on decorating technique where liquid clay (slip) is applied to the surface of a leather-hard piece using a fine-tipped squeeze bottle. The process allows for intricate designs, raised patterns, and textures. Artists can create dots, lines, and shapes, controlling the flow of slip to achieve various effects.

Scope

The scope covers the following :

- Prepare the slip
- Fill the slip trailer
- Test the flow
- Apply the slip
- Touch up and drying

Elements and Performance Criteria

Prepare the Slip

To be competent, the user/individual on the job must be able to:

- PC1.** use the same type of clay that the pottery piece is made from
- PC2.** break the clay into small pieces for easier mixing
- PC3.** Use a hand blender, stick, or even a whisk to blend the clay and water mixture thoroughly
- PC4.** use a fine mesh or sieve to remove any remaining lumps or impurities

Fill the Slip Trailer

To be competent, the user/individual on the job must be able to:

- PC5.** select a slip trailer or squeeze bottle with a nozzle
- PC6.** use a small funnel to carefully pour the slip into the bottle
- PC7.** leave some space at the top (about 1/3 of the bottle empty) to prevent the slip from spilling out when squeezed, and to allow air for smooth flow
- PC8.** lightly squeeze the bottle to ensure the slip flows smoothly through the nozzle

Test the Flow

To be competent, the user/individual on the job must be able to:

- PC9.** Use a scrap piece of clay, a tile, or even a piece of paper to test the flow of the slip before applying it to the final piece
- PC10.** start by applying gentle pressure to the slip trailer or squeeze bottle to see how easily the slip flows
- PC11.** practice making lines, dots, or patterns on the test surface

Apply the Slip

To be competent, the user/individual on the job must be able to:

- PC12.** position the slip trailer close to the surface of your leather-hard clay piece

Qualification Pack

PC13. move the nozzle smoothly over the surface of the clay to create the desired patterns, such as lines, spirals, or dots

PC14. a needle tool can help adjust small mistakes before the slip dries

Touch Up and Drying

To be competent, the user/individual on the job must be able to:

PC15. use a fine needle tool or a damp brush to gently refine the edges or smooth out any imperfections

PC16. give the slip a few minutes to settle on the surface of the clay before moving the piece

PC17. allow the entire piece (both clay and slip) to dry slowly and evenly

PC18. let it dry completely before moving on to further processes like bisque firing or glazing

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. understand the different clay bodies and their properties is essential, as some may react better with slip

KU2. understand how color combinations will appear after firing

KU3. knowledge of design elements like balance, contrast, and pattern is important for creating visually appealing pieces

KU4. familiarity with different motifs and styles can help in developing unique designs

KU5. understand the firing process and how temperatures affect slip and clay body

KU6. knowledge of how different surface treatments (like underglaze or glaze) can influence the application and appearance of slip trails

KU7. Different clay bodies can affect the final appearance of the slip

KU8. different layers will react during firing is essential for achieving the desired look

KU9. clean any excess slip from the base of the pottery before firing to avoid sticking to the kiln shelf

KU10. difference between functional ware and purely decorative pieces is important in choosing the appropriate slip trailing techniques

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. ability to design unique patterns and understand aesthetics

GS2. slip trailing requires precision to ensure that the designs are executed cleanly

GS3. note down the clay properties, slip consistency, and firing temperatures

GS4. knowledge of materials helps in making informed decisions during the process

GS5. knowing how different slips will react during firing and how they interact with the clay body

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare the Slip</i>	8	16	-	-
PC1. use the same type of clay that the pottery piece is made from	2	4	-	-
PC2. break the clay into small pieces for easier mixing	2	4	-	-
PC3. Use a hand blender, stick, or even a whisk to blend the clay and water mixture thoroughly	2	4	-	-
PC4. use a fine mesh or sieve to remove any remaining lumps or impurities	2	4	-	-
<i>Fill the Slip Trailer</i>	8	16	-	-
PC5. select a slip trailer or squeeze bottle with a nozzle	2	4	-	-
PC6. use a small funnel to carefully pour the slip into the bottle	2	4	-	-
PC7. leave some space at the top (about 1/3 of the bottle empty) to prevent the slip from spilling out when squeezed, and to allow air for smooth flow	2	4	-	-
PC8. lightly squeeze the bottle to ensure the slip flows smoothly through the nozzle	2	4	-	-
<i>Test the Flow</i>	4	12	-	-
PC9. Use a scrap piece of clay, a tile, or even a piece of paper to test the flow of the slip before applying it to the final piece	2	4	-	-
PC10. start by applying gentle pressure to the slip trailer or squeeze bottle to see how easily the slip flows	1	4	-	-
PC11. practice making lines, dots, or patterns on the test surface	1	4	-	-
<i>Apply the Slip</i>	4	10	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. position the slip trailer close to the surface of your leather-hard clay piece	1	3	-	-
PC13. move the nozzle smoothly over the surface of the clay to create the desired patterns, such as lines, spirals, or dots	2	4	-	-
PC14. a needle tool can help adjust small mistakes before the slip dries	1	3	-	-
<i>Touch Up and Drying</i>	6	16	-	-
PC15. use a fine needle tool or a damp brush to gently refine the edges or smooth out any imperfections	1	4	-	-
PC16. give the slip a few minutes to settle on the surface of the clay before moving the piece	2	4	-	-
PC17. allow the entire piece (both clay and slip) to dry slowly and evenly	1	4	-	-
PC18. let it dry completely before moving on to further processes like bisque firing or glazing	2	4	-	-
NOS Total	30	70	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	HCS/N0109
NOS Name	Perform Slip Trailing
Sector	Handicrafts and Carpet
Sub-Sector	
Occupation	Production Management-Ceramics
NSQF Level	2.5
Credits	2
Version	1.0
Last Reviewed Date	30/04/2025
Next Review Date	30/04/2028
NSQC Clearance Date	30/04/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score the Recommended Pass % aggregate for the QP.
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Qualification Pack

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Minimum Passing % at NOS Level: 50

(Please note: A Trainee must score the minimum percentage for each NOS separately as well as on the QP as a whole.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
HCS/N0110.Packaging and Presentation of Ceramic Products	30	70	-	-	100	20
HCS/N8509.Entrepreneurship Skill for Ceramics Preparation Artisan	40	60	-	-	100	10
HCS/N9952.Working together as a team	30	70	-	-	100	10
HCS/N9946.Managing workplace safety	30	70	-	-	100	10
HCS/N9937.Maintain Cleanliness	30	70	-	-	100	10
HCS/N9941.Understanding Business Operations	35	65	-	-	100	10
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	10
Total	215	435	-	-	650	80

Elective: 1 Casting Operator- Ceramics

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National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
HCS/N0602.Carry out the processes required to prepare the cast	37	63	-	-	100	20
Total	37	63	-	-	100	20

Elective: 2 Colour Mixing

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
HCS/N0816.Mix the ingredients to obtain the desired colours	32	68	-	-	100	20
Total	32	68	-	-	100	20

Elective: 3 Engobing Operator (Ceramics)

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
HCS/N0703.Engobing of ceramics products	30	70	-	-	100	20
Total	30	70	-	-	100	20

Elective: 4 Furnace Operator

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
HCS/N2104.Carry out furnace operation	25	75	-	-	100	20

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National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
Total	25	75	-	-	100	20

Elective: 5 Glazing Operator

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
HCS/N0817.Perform Glazing of Ceramics Products	20	80	-	-	100	20
Total	20	80	-	-	100	20

Elective: 6 Jigger Operator

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
HCS/N0502.Perform jiggering of Dough	25	75	-	-	100	20
Total	25	75	-	-	100	20

Elective: 7 Moulder

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
HCS/N0408.Prepare the Mould for production	31	69	-	-	100	20
Total	31	69	-	-	100	20

Elective: 8 Mixing Operator

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National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
HCS/N0303.Perform mixing of raw materials using equipment	26	74	-	-	100	10
HCS/N0304.Perform Manual mixing	31	69	-	-	100	10
Total	57	143	-	-	200	20

Elective: 9 Perform Slip Trailing

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
HCS/N0109.Perform Slip Trailing	30	70	-	-	100	20
Total	30	70	-	-	100	20

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.